

SHOUMIK SAHA

Ph.D. Applicant in Computer Science | IEEE & Springer LNNS Published Researcher
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IEEE Xplore: ieeexplore.ieee.org/document/10914487 | Springer LNNS: IJCACI 2026

RESEARCH INTERESTS

NoSQL & Distributed Database Systems · Database Performance Benchmarking · Trustworthy & Explainable AI (XAI) · AI Safety & Governance · Multimodal AI Detection · Human-in-the-Loop Systems · Educational Cybersecurity

EDUCATION

M.Sc. in Computer Science Apr 2023 – Jul 2024

American International University-Bangladesh (AIUB)

- CGPA: 3.73 / 4.00 | Focus: NoSQL database benchmarking, selection frameworks, and document-store performance evaluation

B.Sc. in Computer Science & Engineering Jan 2019 – Dec 2022

American International University-Bangladesh (AIUB)

- CGPA: 3.69 / 4.00 | Specialization: Software Engineering, Databases & Systems Design

PEER-REVIEWED PUBLICATIONS

Khandokar, T., Ahmed, M.M., Sonet, I.H., Bhuiyain Rafi, M.Z., & Saha, S. (2026). SENTRY-AI: A Privacy-Preserving, Human-in-the-Loop Framework for Educational Resilience and Multimodal Detection of AI-Enabled Cyberbullying in U.S. Schools.

Proceedings of IJCACI 2026, Lecture Notes in Networks and Systems (LNNS), Springer — SCOPUS-indexed; accepted from 1,216 submissions

- Co-designed a seven-component multimodal framework (NLP, computer vision, audio analytics, attention-based fusion) for detecting AI-generated cyberbullying in K-12 schools
- Co-designed the privacy-preserving, human-in-the-loop governance architecture (explainable AI via SHAP, audit logging, role-based access control, student appeal pathways) aligned with FERPA, COPPA, CIPA, and NIST AI RMF

Sakib, F.F., Roy, S.D., Rahman, Z., Saha, S., & Salam, A. (2025). NoSQL Database Selection Process: An Integrated Proof of Concept and Comparison for Targeted Document Store Databases.

IEEE International Conference on Advancement in Electrical and Electronic Engineering (ICAEE), IEEE Xplore — DOI: 10.1109/ICAEE

- Designed and executed a multi-dimensional benchmarking study of MongoDB, Couchbase, and CouchDB across performance, scalability, and ACID compliance
- Developed a structured selection framework for document-store NoSQL adoption in enterprise environments

RESEARCH EXPERIENCE

Graduate Research — AI Safety, Cyberbullying Detection & NoSQL Systems, AIUB 2023 – 2026

- Contributed to two internationally accepted research papers spanning database systems (IEEE 2025) and AI governance/educational cybersecurity (Springer LNNS 2026)
- Participated in the full research lifecycle: literature review, framework design, simulation, manuscript writing, peer-review revision, and conference presentation coordination

PROFESSIONAL EXPERIENCE

Software Developer, Selenite Care, Dhaka, Bangladesh May 2026 – Present

- Designing and building a production-ready full-stack web application from the ground up using Next.js, TypeScript, and PostgreSQL, deployed on Vercel with a cloud database backend
- Architecting secure, multi-role authentication and access control with session management and route-level protection across distinct user types
- Integrating multiple international and local payment gateways; building an inventory and order management system supporting multiple fulfillment options
- Developing role-specific admin dashboards with analytics, real-time notifications, and data export; optimizing database performance through indexing, pagination, and caching
- Setting up CI/CD pipelines with automated preview and production deployment

Software Developer (Full-Stack), SRS Software & Tech, Dhaka, Bangladesh Apr 2024 – May 2025

- Designed and developed a production-grade full-stack appointment booking system with real-time availability tracking using the MERN stack (MongoDB, Express.js, React.js, Node.js)

- Architected and optimized MongoDB document schemas for high-throughput query patterns, applying NoSQL performance research directly to production systems

Intern — IT Infrastructure & Database Administration, United Commercial Bank Ltd. (UCBL) 2022

- Administered Oracle databases and executed enterprise backup/disaster recovery operations across banking infrastructure

PROJECTS

BrandSync AI (Sync AI OS) — AI-Powered Marketing & Brand Intelligence Platform 2025

React 19 · TanStack Start · TypeScript · Supabase (Auth + PostgreSQL) · Groq API (LLaMA 3.3-70B) · Firecrawl API

- Co-developed a unified AI-driven platform consolidating audience intelligence, brand reputation monitoring, campaign automation, and brand guideline generation, replacing 10–15 disconnected MarTech tools
- Implemented the full authentication system (email/password + Google OAuth) and integrated Google Analytics 4, Search Console, and YouTube APIs via OAuth 2.0
- Designed a 12+ table PostgreSQL schema with row-level security (RLS); built server-side functions powering the Audience, Reputation, and Campaign modules
- Live application: sync-ai-os.lovable.app

AI Developer Assistant Agent — AI-Assisted Repository Analysis Tool (In Progress) 2026

TypeScript · Node.js · LangChain · GitHub REST API · Google Gemini API

- Building an independent agent that retrieves GitHub repository metadata via the REST API and generates LLM-based architectural summaries using LangChain and Gemini
- Exploring AI-assisted software engineering; roadmap includes source-level analysis and automated technology-stack detection

Selenite Care — Membership-Based Skincare Consultation & E-commerce Platform 2025

selenitecare.com

- Developed and deployed a full-stack healthcare e-commerce platform using Next.js 16, TypeScript, Tailwind CSS, PostgreSQL, Prisma ORM, and Supabase, with production deployment on Vercel
- Designed and implemented a secure multi-role authentication and authorization system (Client, Doctor, CRM, Admin) using NextAuth v5, JWT sessions, email verification, and role-based access control
- Engineered a comprehensive membership and appointment management system, including automated membership lifecycle management, doctor availability scheduling, and digital skin consultation workflows with image uploads and AI-assisted product recommendations
- Integrated Stripe and bKash payment gateways, built complete order and inventory management modules, and implemented automated transactional email services using Resend
- Architected scalable backend infrastructure with 20+ Prisma database migrations, production-ready CI/CD using GitHub Actions and Vercel, real-time dashboards, analytics, notifications, and CSV reporting for administrative operations

ACHIEVEMENTS

Finalist, The Infinity AI Buildfest — National competitive technology event, Bangladesh

- Reached the finalist stage as a full-stack developer on the BrandSync AI team
- Verification: cloudfcampbd.com/verify/4111729b5282737cc45ef8bc

TECHNICAL & METHODOLOGICAL SKILLS

- **Languages:** JavaScript, TypeScript, PHP, Python
- **Frameworks:** React.js, Next.js, Node.js, Express.js, Laravel
- **Databases:** MongoDB, Couchbase, CouchDB, MySQL, PostgreSQL, Oracle — benchmarking, schema design, performance optimization
- **AI/ML Tools:** scikit-learn, SHAP (XAI), Transformers (BERT, RoBERTa), RandomForest
- **Research Methods:** Experimental design, POC development, performance benchmarking, comparative analysis
- **Dev Tools:** Git, Docker, Vercel, Railway, Redux Toolkit

LANGUAGES & ENGLISH PROFICIENCY

- English (Fluent) — IELTS Academic: Overall 6.5 (Writing 7.0, Listening 6.5, Speaking 6.5, Reading 5.5); B.Sc. and M.Sc. conducted entirely in English (MOI Certificate available)
- Bengali — Native speaker